

(a) Shared executed prefix

Commitments C + deliberate releases E
fixed base B; compatible pool R

Example response interactions



Executed warm: $L = 8$

Clique bound: $U = \max\{8, 8, 6\} + 6 = 14$

(b) Joint residual prediction

Shared encoder for observable (R, W)
rewards · conflicts · resource factors

Sparse factor readout
 $z \rightarrow p$, with $\sum_{v \in Q} p_v \leq 1$

Request head
native workpoint + remaining time

$$\hat{f} = L + (U - L) \cdot \text{sigmoid}(f)$$
$$\hat{g} = q + \hat{f}$$

(c) Budgeted improvement

Prioritize unspent requests
 $[\hat{g} - \text{best gain}]_+ / \text{expected cost}$

Execute finite native search
use the same actual W for every policy

Check original edges + objective
accept only timely improvements

Report shared prefix and extra gain
strict return time includes full validation

Offline only

Same actual state
and common warm W

Clone → execute alternatives
full-budget value / returned mask

Executed-value ranking + mask supervision
train the joint residual model

All methods pay the same prefix; Greedy-only stops there. The 4-node example is mathematical, not an experimental outcome.